

Wittle Shop Project

Wittle Shop (Pty) Ltd, is a South African online retail company that sells electronics, clothing and home appliances directly to customers through a web and mobile platform. The company is growing fast and they want to build a new database system to manage their customers, products, orders, payments, deliveries and supplies. In this regard, they engaged my services to design their database structure.

Given the Business requirements they submitted, I started by brainstorming ideas on the entities and their relationships in the database. I then utilized draw.io diagramming tool to create entity tables and indicate their relationship.

1. The Entities for the database are as follows;

No.	Entities	Purpose
1.	Customer	To register buyers
2.	Address	For Delivery addresses
3.	Category	Product groupings
4.	Supplier	Companies that supply products
5.	Product	Items for sale
6.	Order	Orders placed by customers
7.	Order item	Items placed for order
8.	Payment	Payment for an order
9.	Delivery	Delivery of an order to an address

2. The Primary and Foreign Keys from all tables were identified as follows;

No.	Entities	Primary Key	Foreign Key
1.	Customer	Customer ID	N/A
2.	Address	Address ID	Customer ID
3.	Category	Category ID	N/A
4.	Supplier	Supplier ID	N/A
5.	Product	Product ID	Supplier ID
6.	Order	Order ID	Product ID, Customer ID
7.	Order item	Order ID, Product ID	Order ID, Product ID
8.	Payment	Payment ID	Customer ID
9.	Delivery	Delivery ID	Address ID

3. Relationships and Cardinality

No.	Relationship	Cardinality	Explanation
1.	Customer-Address	1: M	A customer can have multiple delivery addresses
2.	Customer-Order	1: M	A customer can place multiple orders.
3.	Category -Product	1: M	Each product belongs to one category; A category can have many products
4.	Supplier- Product	1: M	Each product is supplied by one supplier; A supplier can supply many products

5.	Order-Product	M: N	An order can contain multiple products; A product can appear in multiple orders
6.	Order-Order Item	1: M	One order has many line items
7.	Product-Order Item	1: M	One product appears on many line items
8.	Order-Payment	1: 1	One order → one payment
9.	Order-Delivery	1: 1	One order → one delivery
10.	Address-Delivery	1: M	One address can be used for many deliveries; each delivery goes to one address